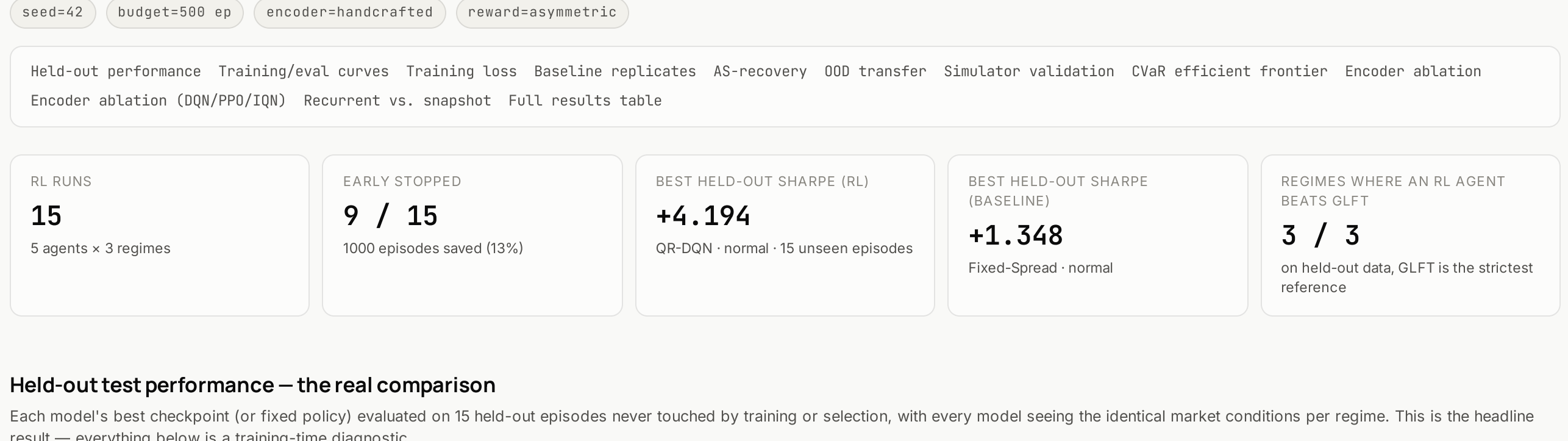


# DISRL Market Maker – Training & Evaluation Results

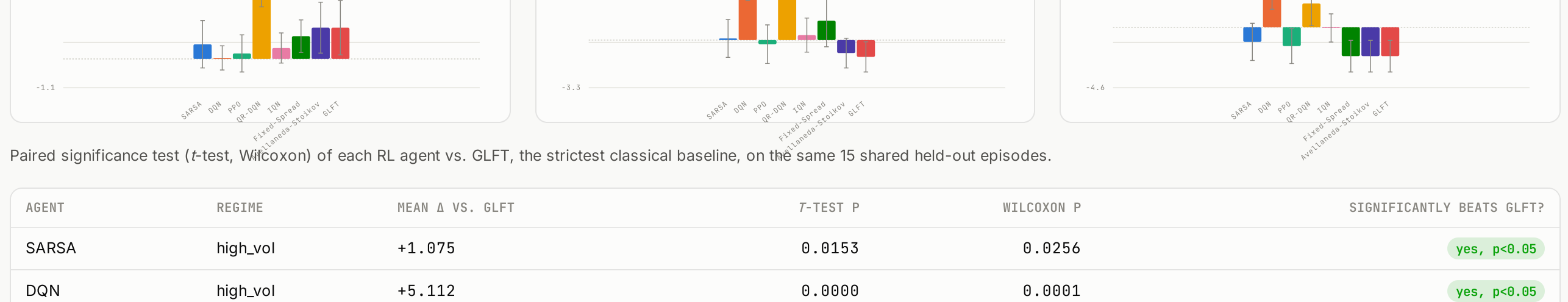
5 RL agents vs. 3 analytical baselines across 3 volatility regimes in an ABIDES-Gym limit-order-book simulator, plus out-of-distribution transfer, CVaR risk-sensitivity, and state-representation ablations.



## Hold-out test performance – the real comparison

Each model's best checkpoint (or fixed policy) evaluated on 15 held-out episodes never touched by training or selection, with every model seeing the identical market conditions per regime. This is the headline result – everything below is a training-time diagnostic.

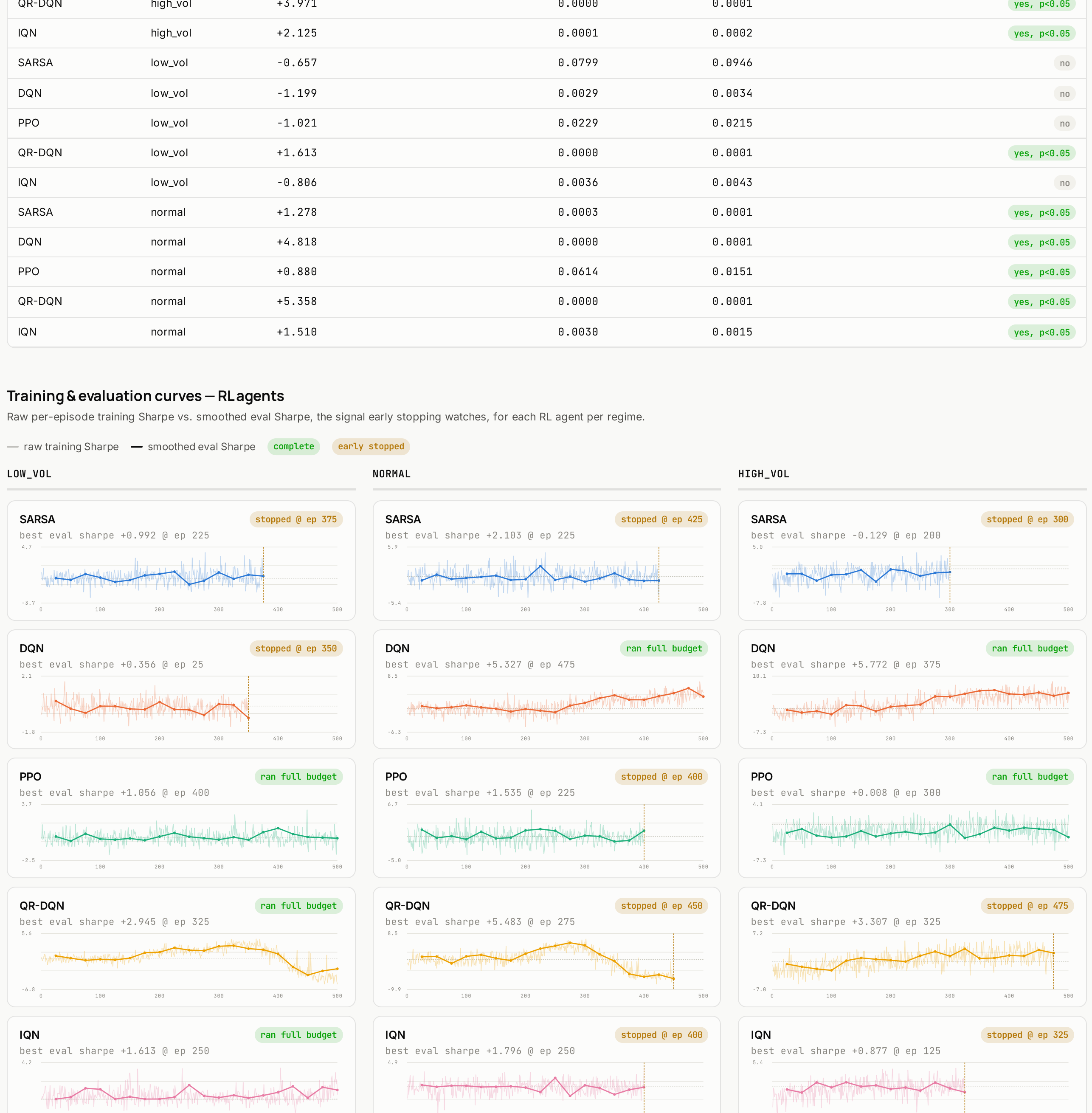
SARSA | DQN | PPO | QR-DQN | IQN | Fixed-Spread | Avelaneda-Stoikov | GLFT



## Training & evaluation curves – RL agents

Raw per-episode training Sharpe vs. smoothed eval Sharpe, the signal early stopping watches, for each RL agent per regime.

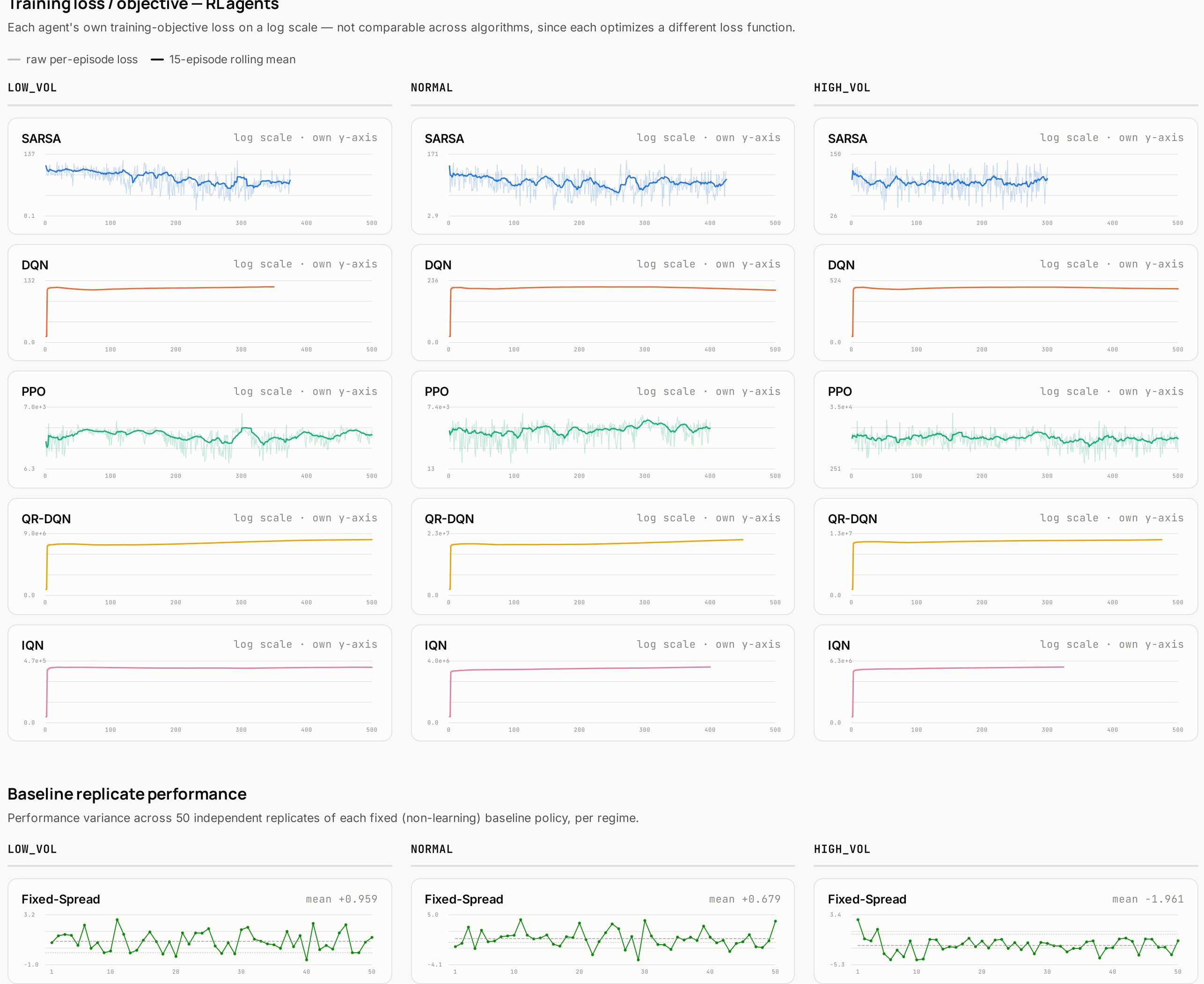
— raw training Sharpe — smoothed eval Sharpe complete early stopped



## Training loss / objective – RL agents

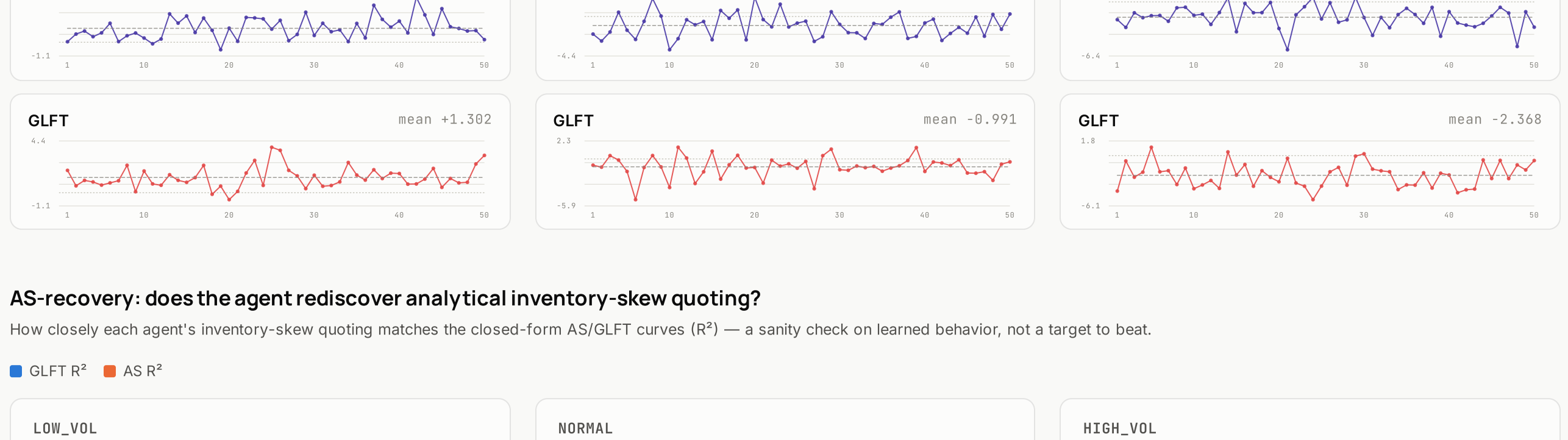
Each agent's own training-objective loss on a log scale – not comparable across algorithms, since each optimizes a different loss function.

— raw per-episode loss — 15-episode rolling mean



## Baseline replicate performance

Performance variance across 50 independent replicates of each fixed (non-learning) baseline policy, per regime.



## AS-recovery: does the agent rediscover analytical inventory-skew quoting?

How closely each agent's inventory-skew quoting matches the closed-form AS/GLFT curves (R<sup>2</sup>) – a sanity check on learned behavior, not a target to beat.

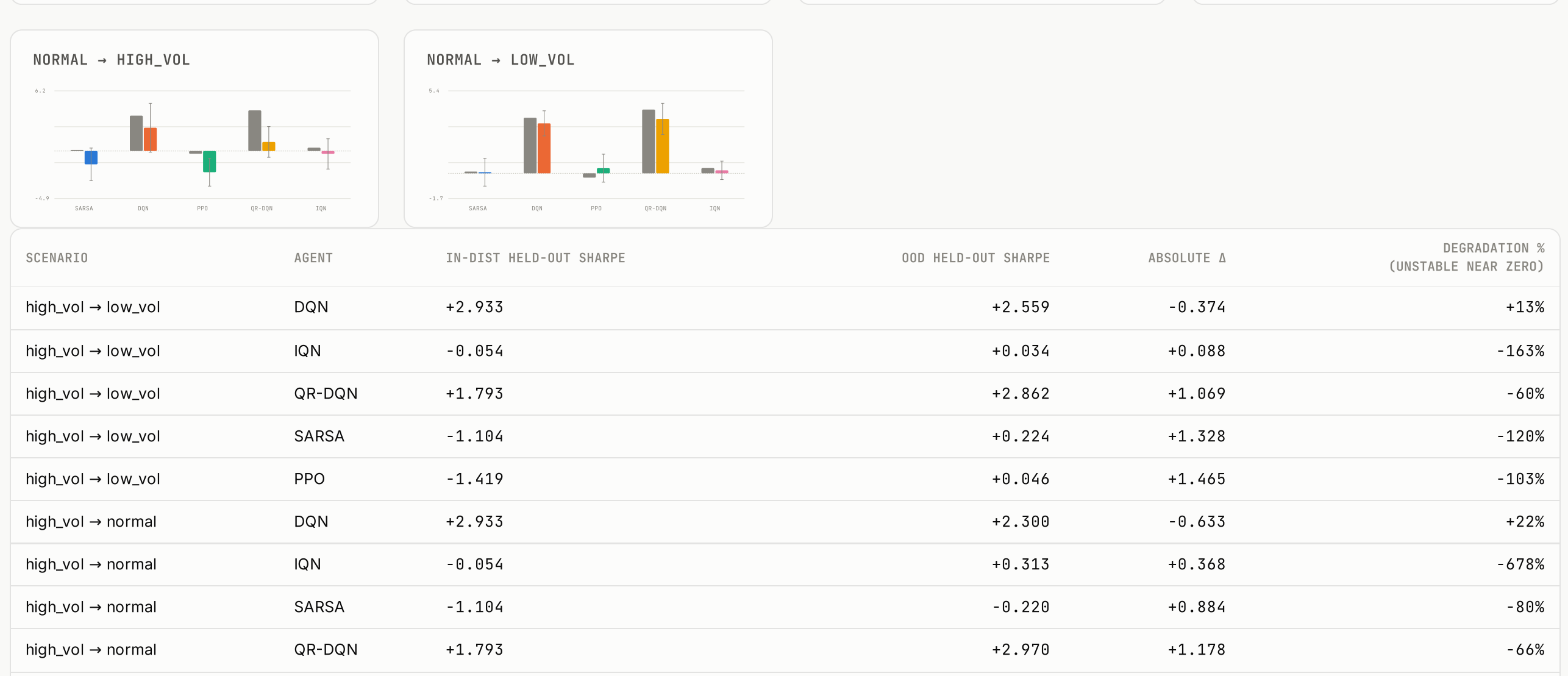
■ GLFT R<sup>2</sup> ■ AS R<sup>2</sup>



## OOD transfer: agents evaluated outside their training regime

Each agent's dropped from one regime dropped into a different regime with no fine-tuning, across all 6 directed pairs among low\_vol/normal/high\_vol – tests generalization to market conditions never seen in training. QR-DQN stays profitable in all 6 of 6 transfers and actually improves on its in-distribution Sharpe in 3 of them; transferring into high\_vol is the hardest direction for most agents.

■ in-distribution ■ out-of-distribution



## Simulator validation – stylized facts

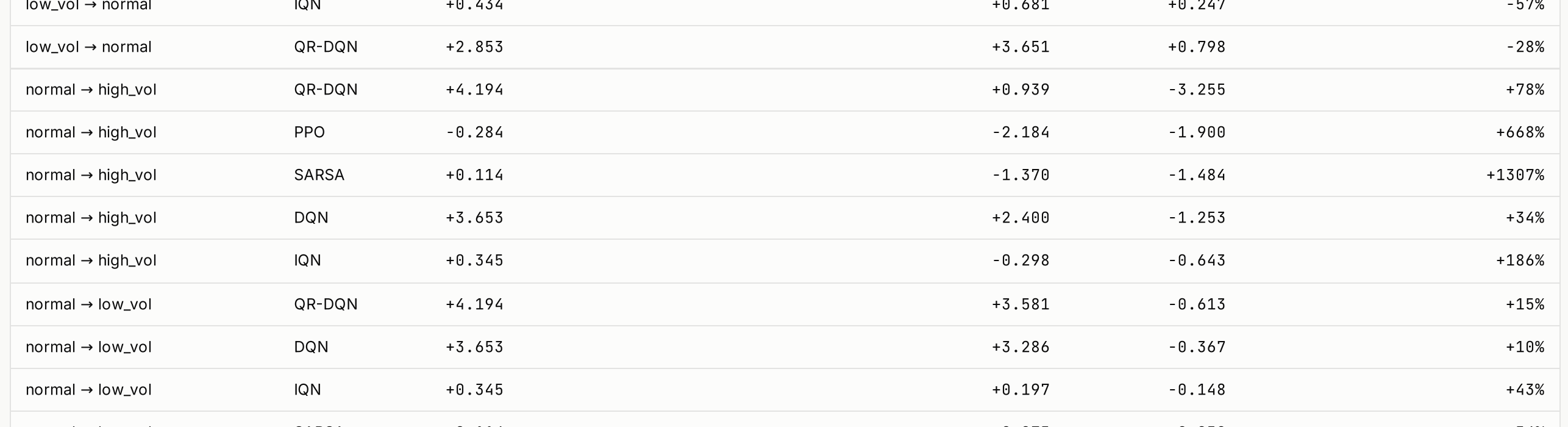
A random-policy audit of the simulator itself against 5 standard empirical market-microstructure regularities, independent of any trained agent – validates that the environment produces realistic dynamics to learn from.



## CVaR efficient frontier – QR-DQN&IQN, normal regime

Mean P&L vs. tail risk (CVaR<sub>0.95</sub>) across the CVaR-α sweep, averaged over 4 training seeds per setting. Most α values are statistically indistinguishable once seed variance is accounted for – CVaR-α doesn't produce the smooth risk/return tradeoff the method is meant to deliver at this training budget.

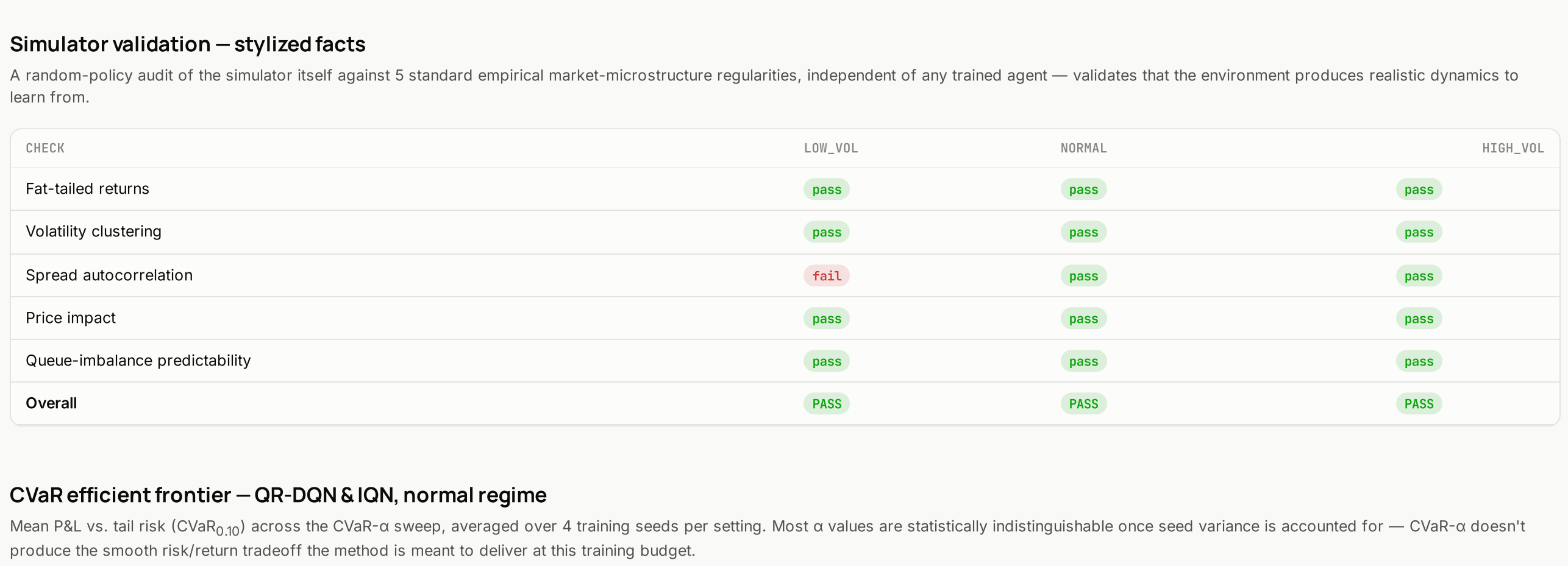
— mean held-out P&L (±1 std, 4 seeds) — CVaR<sub>0.95</sub> (±1 std, 4 seeds)



## Encoder ablation – QR-DQN, normal regime

QR-DQN's held-out Sharpe using handcrafted features vs. two learned snapshot representations (CNN, autoencoder) – handcrafted wins decisively (+4.19 vs. +0.45 and -0.18, both p<0.0001).

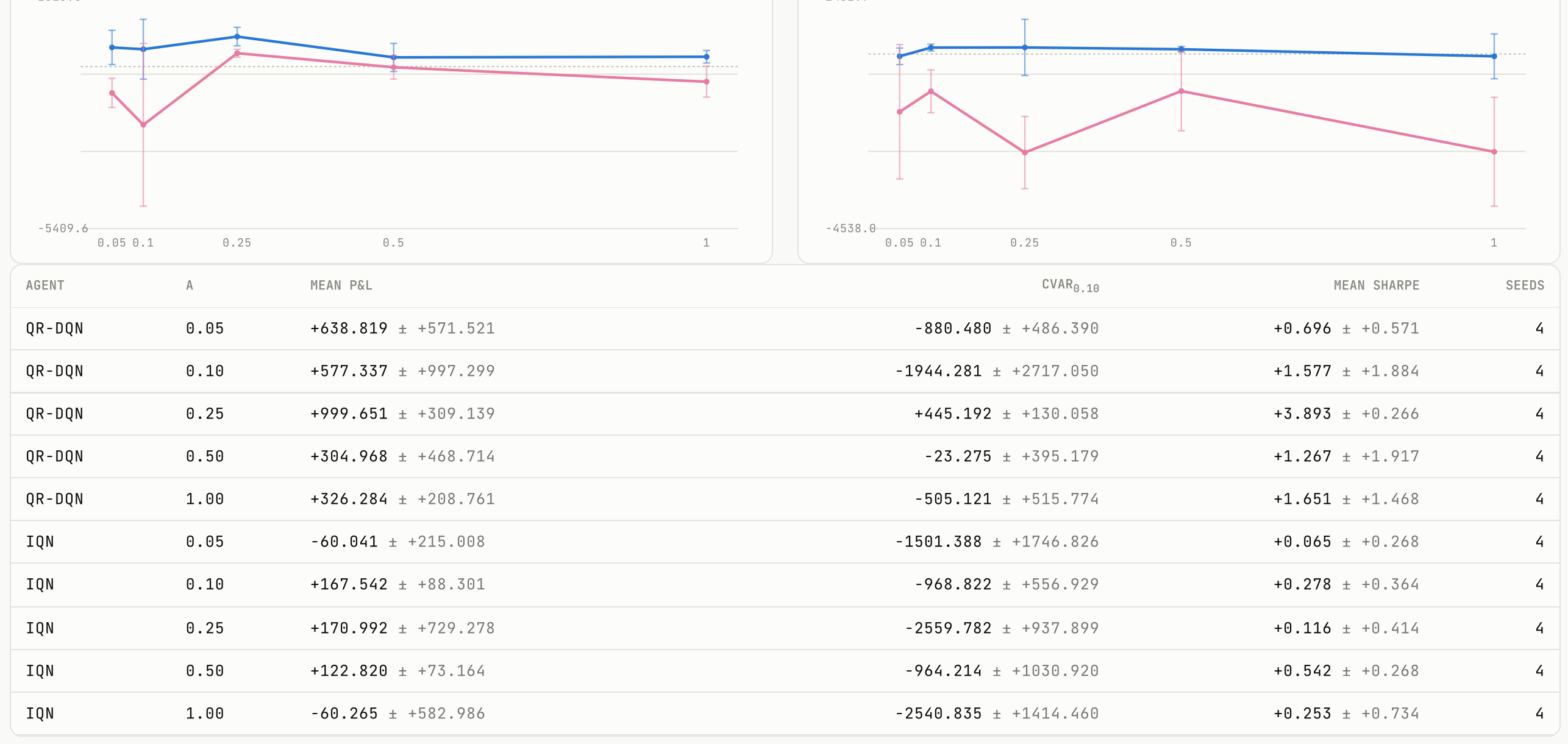
■ Handcrafted ■ CNN ■ Autoencoder



## Recurrent vs. snapshot: does temporal memory help?

Each agent's standard snapshot-and-encoder policy vs. the same agent with an LSTM backbone in its core, across 10 agent/regime/CVaR-risk combinations. Recurrent integration never significantly helps, significantly hurts QR-DQN and DQN in the normal regime, and its effect when combined with CVaR risk-sensitivity is alpha-dependent rather than a consistent interaction.

■ snapshot ■ recurrent (LSTM)



## Full results table

Every figure charted above, in one place.

QR-DQN's held-out Sharpe using handcrafted features vs. two learned snapshot representations (CNN, autoencoder) — handcrafted wins decisively (+4.19 and +0.45 and -0.38, both p<0.0001).

Handcrafted

CNN

Autoencoder

QR-DQN HELD-OUT SHARPE BY ENCODER

6.0